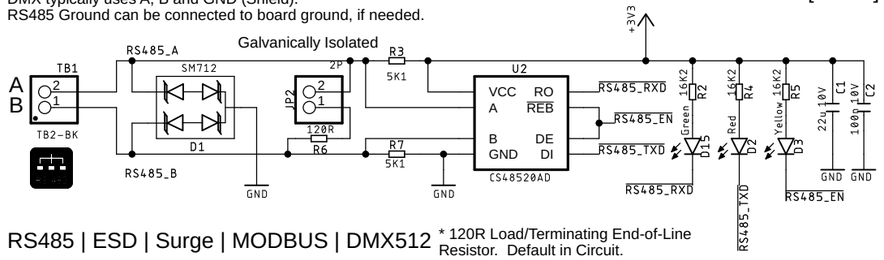
[UART]

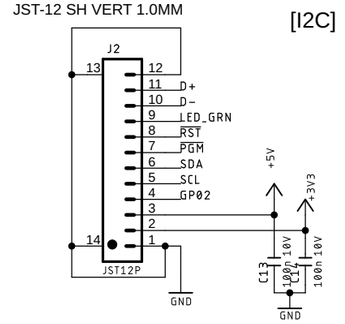
RS485 typically only uses A and B. Unless shielded cables are used.
DMX typically uses A, B and GND (Shield).
RS485 Ground can be connected to board ground, if needed.



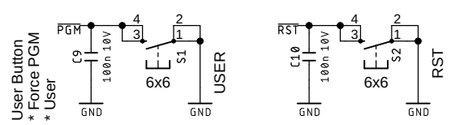
RS485 | ESD | Surge | MODBUS | DMX512 * 120R Load/Terminating End-of-Line Resistor. Default in Circuit.

[I2C]

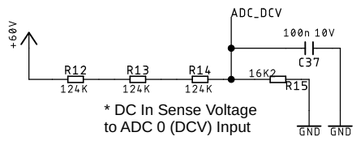
JST-12 SH VERT 1.0MM



Expansion Interface | OLED (3V3)



Tactile Switches | Reset | Usr/Pgm



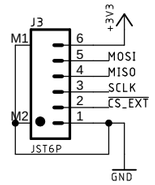
Appx DCV/ADC

05 V	0.17 V
12 V	0.40 V
20 V	0.67 V
30 V	1.01 V
40 V	1.35 V
50 V	1.69 V
60 V	2.03 V
70 V	2.37 V
75 V	2.54 V

ADC DC Volts

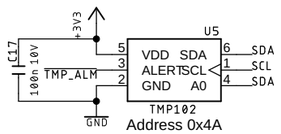
[SPI]

JST-6 SH VERT 1.0MM



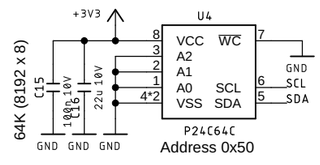
SPI Expansion | MicroSD (3V3)

[I2C]



Temperature | TMP102

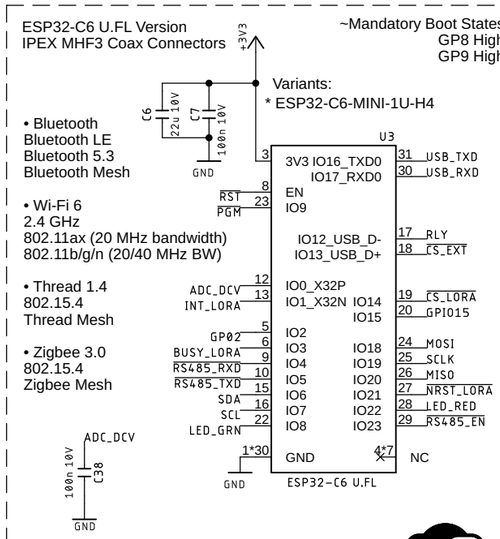
[I2C]



EEPROM | 24C64

ESP32-C6 U.FL Version
IPEX MHF3 Coax Connectors

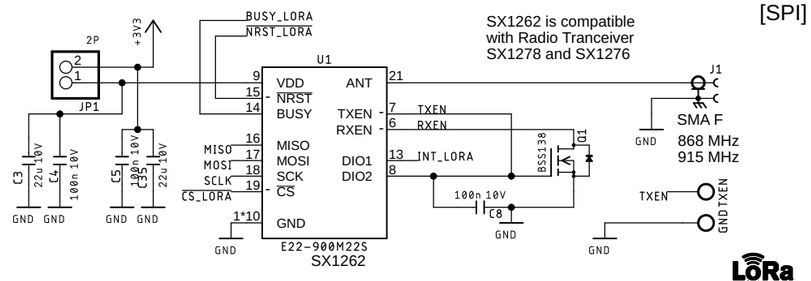
~Mandatory Boot States
GP8 High
GP9 High



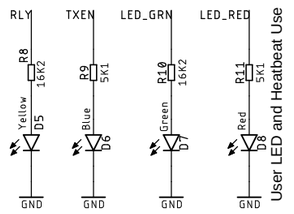
MCU | ESP32-C6

ESPKite
LoRa Controller

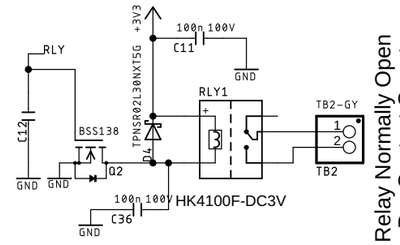
[SPI]



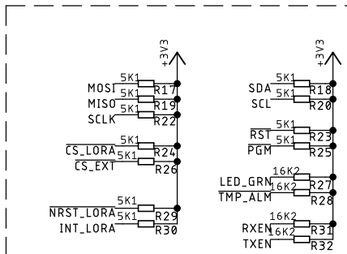
LoRa | EByte E22-900M22S | SX1262



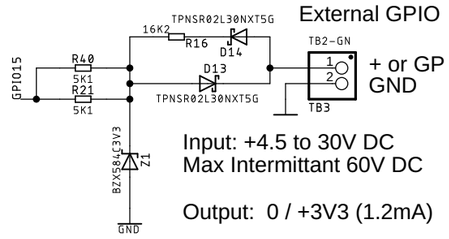
Status | LED



Relay Output (Galvanically Isolated)



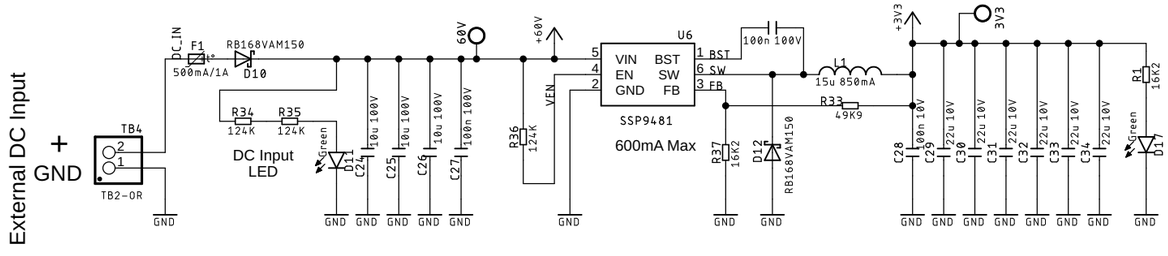
GPIO Pullups



GPIO15 | GPIO Input or Output

Beware of any High Voltages

DC In 5-60V. Maximum 75V Input.
Permanant damage WILL occur > 75V !
* Reverse Polarity Protected
* Fuse Hold 500mA Trip 1A Resettable

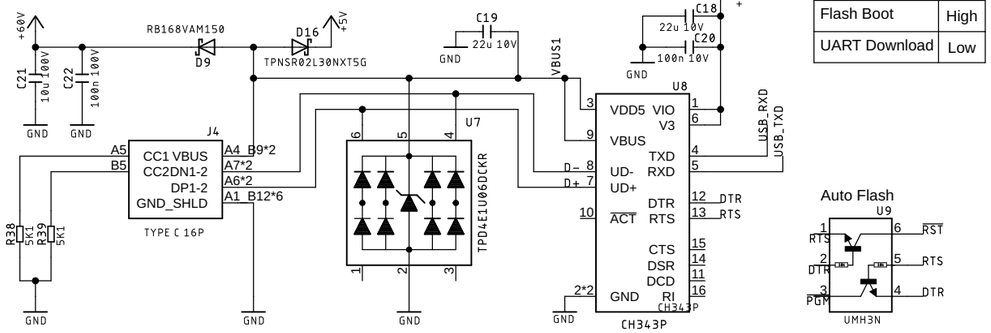


High Voltage Power Suppy | HVDC SMPS 3V3 DC-DC

USB UART providing Arduino/PlatformIO Flashing ESP32-C6 DEV MODULE Compatibility
Board compatible with esp32-c6-devkitm-1 and adafruit_feather_esp32c6

GPIO9 + Reset

Flash Boot	High
UART Download	Low



USB | USB UART | Serial Monitoring | USB Flashing | Debug | ESD | 5V Power

Enclosure VG-P41F 80x54x32 Main PCB 79x49mm

For further information and code examples, please see our code and GitHub WIKI pages. ditronix.net | GitHub.com/DitroniX | Hackster.io/DitroniX

Components optimised for BOM

DitroniX
Home Automation | Internet of Things | Energy Monitoring

Production

(C)2021-2026 - DitroniX - Dave Williams G8PUO

STEM

ESP32-C6 SX1262 SDK **ditronix.net** DitroniX Ltd – UK	ESPKite ESP32C6 E22-900M22S SX1262 1.2605.102	
	ESPKITE-LORA-ESP32-C6-RS485-CONTROLLER-SDK	
Sheet: 1/1	Drwg/Chk: **Dave Williams**	18/05/2026 18:15